

# Inhyeok Choi

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School of Mathematics, Korea Institute for Advanced Study  
85 Hoegi-ro, Dongdaemun-gu, Seoul 02455, South Korea  
Email: [inhyeokchoi48@gmail.com](mailto:inhyeokchoi48@gmail.com)  
Website: <https://inhyeokchoi48.github.io>

## Research Interests

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My research lies in the intersection of geometric topology, geometric group theory and probability theory. Research keywords include Teichmüller space, mapping class groups, Outer space,  $\text{Out}(F_N)$ , Gromov hyperbolic spaces,  $\text{CAT}(0)$  spaces, random walks, growth problems and counting problems in groups, percolations in vertex-transitive graphs, and dynamics of groups acting on spaces.

## Positions

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| 08/2025–        | <b>Korea Institute for Advanced Study</b> , Seoul, South Korea<br>Professor (07/2026–), KIAS Fellow (–06/2026)                                                                     |
| 07/2024–08/2025 | <b>Cornell University</b> , Ithaca, NY, United States<br>Harry Kesten Assistant Professor (Post-doc)                                                                               |
| 03/2023–06/2024 | <b>Korea Institute for Advanced Study</b> , Seoul, South Korea<br>June E Huh Fellow (07/2023–), Project research fellow (–06/2023) (Post-doc)<br>Mentor: Sang-hyun Kim             |
| 03/2024–06/2024 | <b>Fields Institute</b> , Toronto, ON, Canada<br>Marsden Postdoctoral Fellowship (visiting for the program “Thematic program on Randomness and Geometry”)<br>Mentor: Giulio Tiozzo |

## Education

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| 09/2018–02/2023 | <b>KAIST</b> , Daejeon, South Korea<br>Ph.D. in Mathematical Sciences<br>Advisor: Hyungryul Baik                     |
| 03/2015–08/2018 | <b>KAIST</b> , Daejeon, South Korea<br>B.Sc. in Physics/Mathematical Sciences/Biological Sciences<br>Summa Cum Laude |

## Other Educational Experiences

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| 01/2026–05/2026 | Postdoc at Simons Laufer Mathematical Sciences Institute                                |
| 07/2024–08/2025 | June E Huh Visiting fellow, HCMC of KIAS                                                |
| 12/2018–02/2019 | Exchange student, Tokyo Institute of Technology                                         |
| 08/2017–02/2018 | Exchange student, École Polytechnique Fédérale de Lausanne                              |
| 12/2015–06/2017 | Undergraduate researcher, <b>Biomedical Optics Lab, KAIST</b><br>Advisor: YongKeun Park |
| 06/2015–08/2015 | Exchange student, University of California, Berkeley                                    |

## Awards and Honors

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|---------|----------------------------------------------------------------------------|
| 09/2025 | Sangsan Young Mathematician Award, the Korean Mathematical Society         |
| 12/2024 | Dissertation of the Year, S-OIL Science Prodigy and Culture Foundation     |
| 02/2023 | Best Thesis Award, 2023, Dept. of Mathematical Sciences, KAIST             |
| 08/2022 | Best TA Award, 2022 Spring semester, Dept. of Mathematical Sciences, KAIST |

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|-----------------|--------------------------------------------------------------------------------------------------------|
| 03/2017         | Dean's list, 2016 Fall semester, College of Natural Sciences, KAIST                                    |
| 12/2016         | Silver Prize, Korean Undergraduate Mathematics Competition (전국 대학생수학경시대회), Korean Mathematical Society |
| 03/2015–08/2018 | Korea Presidential Science Scholarship (대통령과학장학생), Korea Student Aid Foundation                        |
| 03/2015–08/2018 | KAIST Presidential Fellowship                                                                          |

## Publications & Preprints

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1. Classification of horospherical invariant measures in higher rank: The Full Story  
With *Dongryul M. Kim*.  
Preprint (2026), arXiv:2601.22668.
2. Divergence-type semigroups in Kleinian groups and Hausdorff dimension  
Preprint (2025), arXiv:2512.02829.
3. Invariant measures on the space of measured laminations for subgroups of mapping class group  
With *Dongryul M. Kim*.  
**Inventiones Mathematicae** (2026) <https://link.springer.com/article/10.1007/s00222-026-01454-0>.
4. Percolation in acylindrically hyperbolic groups  
With *Donggyun Seo*.  
Preprint (2025), arXiv:2508.08932.
5. Metric weak proper discontinuity for pseudo-Anosov maps  
Preprint (2025), arXiv:2506.21214.
6. Acylindrically hyperbolic groups and counting problems  
Preprint (2025), arXiv:2504.20985.
7. Pivoting technique for the circle homeomorphism group  
Preprint (2025), to appear in **Communications in Mathematical Physics**. arXiv:2504.02788.
8. Smoothing countable group actions on metrizable spaces  
With *Sang-hyun Kim*.  
Preprint (2024), to appear in the **Proceedings of the Mathematical Society of Japan, Seasonal Institute**. arXiv:2410.06077.
9. Counting pseudo-Anosovs as weakly contracting isometries  
**Inventiones Mathematicae**, Vol. 242, pp. 337–386. (2025) <https://doi.org/10.1007/s00222-025-01358-5>.
10. Confined subgroups in groups with contracting elements  
With *Ilya Gekhtman, Wenyuan Yang and Tianyi Zheng*.  
Preprint (2024). arXiv:2405.09070.
11. Contracting isometries and differentiability of the escape rate  
**Probability Theory and Related Fields**, Vol. 194, pp. 1055–1122. (2026) <https://doi.org/10.1007/s00440-025-01434-0>
12. Random walks on groups and superlinear divergent geodesics  
With *Kunal Chawla, Vivian He and Kasra Rafi*.  
**Ergodic Theory and Dynamical Systems**, Vol. 45, no. 5, pp. 1403–1443. (2025) <https://doi.org/10.1017/etds.2024.75>.
13. Genericity of contracting geodesics in groups  
With *Kunal Chawla and Giulio Tiozzo*.  
Preprint (2023). arXiv:2308.01877. **Groups, Geometry, and Dynamics**, online first version. <https://ems.press/journals/ggd/articles/14298985>.

14. Random walks and contracting elements III: Outer space and outer automorphism group  
Preprint (2023). arXiv:2212.12122.
15. Random walks and contracting elements II: Translation lengths and quasi-isometric embedding  
**Groups, Geometry, and Dynamics**, Vol. 19, pp. 1373-1423. (2025) <https://doi.org/10.4171/ggd/831>.
16. Random walks and contracting elements I: Deviation inequality and limit laws  
**Compositio Mathematica**, Vol. 161, pp. 1512-1575. (2025) <https://doi.org/10.1112/S0010437X25007511>.
17. Random walks on mapping class groups (survey paper)  
With *Hyungrul Baik*.  
**EMS Surveys in Mathematical Sciences**, Vol. 9, no. 2, pp. 279–320. (2022) <https://doi.org/10.4171/EMSS/59>.
18. Pseudo-Anosovs are exponentially generic in mapping class groups  
**Geometry & Topology**, Vol. 28, pp. 1923-1955. (2024) <https://doi.org/10.2140/gt.2024.28.1923>.
19. Central limit theorem and geodesic tracking on hyperbolic spaces and Teichmüller spaces  
**Advances in Mathematics**, Vol. 431, 109236. (2023) <https://doi.org/10.1016/j.aim.2023.109236>.
20. Linear growth of translation lengths of random isometries on Gromov hyperbolic spaces and Teichmüller spaces  
With *Hyungrul Baik and Dongryul M. Kim*.  
**Journal of the Institute of Mathematics of Jussieu**, Vol. 23, no. 4, pp. 1751–1795. (2023) <https://doi.org/10.1017/S1474748023000373>.
21. Simple length spectra as moduli for hyperbolic surfaces and rigidity of length identities  
With *Hyungrul Baik and Dongryul M. Kim*.  
Preprint (2020). arXiv:2012.05652. To appear in **Annales de l’Institut Fourier**.
22. On the surjectivity of the Symplectic representation of the mapping class group  
With *Hyungrul Baik and Dongryul M. Kim*.  
**Topology and its Applications**, Vol. 322, 108334. (2022) <https://doi.org/10.1016/j.topol.2022.108334>.
23. Topological entropy of pseudo-Anosov maps from a typical Thurston construction  
With *Hyungrul Baik and Dongryul M. Kim*.  
**International Mathematics Research Notices**, Vol. 2022, no. 24, pp. 19762–19904. (2022) <https://doi.org/10.1093/imrn/rnab167>
24. Inhyeok Choi, KyeoReh Lee, and YongKeun Park. Compensation of aberration and speckle noise in quantitative phase imaging using lateral shifting and spiral phase integration. **Optics Express**, 25(24) pp. 30771–30779. (2017)

## Research Talks

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### Lecture series

- 2024.07.29. Growth of (sub)groups with hyperbolicity. (3 lectures and 1-week group discussion)  
*Geometry in Groups 2024. TIFR-ICTS, Bangalore, India.*
- 2024.05.03. Hyperbolicity of geodesics, counting and random walks. (4 lectures) *Riverside Workshop on Geometric Group Theory 2024, UC Riverside.*

### Invited Talks

- 2026.09.04. Counting curves for subgroups of MCG. *Thematic Programme on Surface Group Representation and Analytic Group Theory*. Lodha Mathematical Sciences Institute, Mumbai, India.
- 2026.07.27. Counting curves for subgroups of MCG. *2K-GATE Workshop III: Groups, dynamics and surfaces on Jeju island*. Jeju, South Korea.
- 2026.05.22. Understanding groups beyond hyperbolicity. *"Current Challenges in Topology": Samsung Global Research Symposium*. Milan, Italy.
- 2026.05.04. Metric weak proper discontinuity for pseudo-Anosov maps. *Geometric Group Theory Seminar*. Purdue University, IN, USA.
- 2026.04.30. Genericity of pseudo-Anosovs in the mapping class group. *Geometry & Topology Seminar*. University of Chicago, IL, USA.
- 2026.04.27. Invariant measures for subgroups of the mapping class group. *Dynamics Seminar*. University of Chicago, IL, USA.
- 2026.04.08. The growth of confined subgroups on a hyperbolic space. *RTG Seminar on Geometry, Dynamics and Topology*. University of Michigan, USA.
- 2026.03.11. Random subgraphs of the mapping class group. *McGill Geometric Group Theory seminar*. McGill University, Montréal, Quebec, Canada.
- 2026.02.24. Random subgraphs of the mapping class group. *OSU Topology Seminar*. Ohio State University, Columbus, OH, USA.
- 2026.02.18. Contracting properties and their dynamical consequences. *TGS Seminar at SLMath*. CA, USA.
- 2026.02.17. Counting problems on hyperbolic groups and beyond. *UCSB Topology Seminar*. CA, USA.
- 2026.01.20. Regularity of the escape rate and the asymptotic entropy for negatively curved groups. *Probability Workshop in Korea 2026*. Jeonju, South Korea.
- 2026.01.12. Invariant measures for subgroups of mapping class groups and Anosov groups. *Perspectives in Ergodic theory, Randomness and Geometry*. Jagiellonian University, Kraków, Poland.
- 2026.01.06. Exponential synchronization on the circle. *France-Korea workshop on Dynamical Group Theory*. Aix-Marseille Université, France.
- 2025.11.19. Statistics in non-positively curved groups. *KIAS Math-HCMC Fall Workshop 2025*. Jeongseon, South Korea.
- 2025.10.14. Locally finite invariant measures on measured laminations. *KIAS conference on mapping class groups*. KIAS, South Korea.
- 2025.09.29. Nonuniqueness from percolation in acylindrically hyperbolic groups. *Alfred Renyi Insitute*, Budapest, Hungary.
- 2025.09.22. Nonuniqueness from percolation in acylindrically hyperbolic groups. *Autumn School on Probability and Dynamics on Groups*, University of Münster, Germany.
- 2025.09.17. Metric WPD of pseudo-Anosov maps. *Groups, Dynamics and Surfaces in Paris*. IMJ-PRG at Sorbonne, Paris, France.
- 2025.08.27. Many infinite clusters from percolation in many groups. *KIAS-KAIST Workshop on current challenges in mathematics*. KIAS, South Korea.
- 2025.08.18. Metric WPD of pseudo-Anosov maps and quasimorphisms. *Algebraic and analytic methods in group theory in Warsaw*. IMPAN, Warsaw, Poland.
- 2025.08.06. Dynamics of semigroups of circle homeomorphisms. *One-day Festival of Group actions, Geometry and Dynamics*. KIAS, South Korea.
- 2025.06.16. Genericity problem in groups with negative curvature. *Geometric Group Theory and Related Topics*. BICMR, Peking University, China.

- 2025.05.19. Counting pseudo-Anosovs and fully irreducibles. *53rd Barrett Lectures: Recent Development in Geometric Group Theory*. TN, USA.
- 2025.04.15. Genericity of pseudo-Anosovs and quasi-isometries. *Stanford Topology Seminar*. CA, USA.
- 2025.04.01. Genericity of pseudo-Anosovs and quasi-isometries. *Geometry & Topology Seminar*. Rutgers New Brunswick, NJ, USA.
- 2025.03.14. Dynamics of semigroups of circle homeomorphisms. *The Scottish Dynamics Network*. Heriot-Watt University, Edinburgh, UK.
- 2025.03.12. Genericity of pseudo-Anosovs and quasi-isometries. *Algebra, Geometry, and Topology Seminar*. Heriot-Watt University, Edinburgh, UK.
- 2025.03.06. Genericity of pseudo-Anosovs and quasi-isometries. *Binghamton Geometry & Topology Seminar*. NY, USA.
- 2025.02.14. Genericity of pseudo-Anosovs and quasi-isometries. *SUNY at Buffalo Geometry & Topology Seminar*. NY, USA.
- 2024.01.16. Free discrete subgroups of  $\text{Homeo}(S)$  and the fine curve graph. *Virtual Seminar in Geometry and Topology (VISGAT)*, KIAS.
- 2024.12.03. Metric WPD for  $\text{Homeo}(S)$ : loxodromics on the fine curve graph. *KAIST Topology Seminar*.
- 2024.11.28. Metric WPD for  $\text{Homeo}(S)$ : loxodromics on the fine curve graph. *Hyperbolic Lunch*, University of Toronto.
- 2024.10.22. Genericity of pseudo-Anosovs and quasi-isometries. *CUNY Geometry & Topology Seminar*, CUNY.
- 2024.10.15. Genericity of pseudo-Anosovs and quasi-isometries. *Geometry & Topology Seminar*, Yale University.
- 2024.10.14. Growth rate of confined subgroups of a discrete group. *Group Action and Dynamics Seminar*, Yale University.
- 2024.09.24. Genericity of pseudo-Anosovs and quasi-isometries. *KIAS-Rice Workshop on Geometric Topology*, KIAS HCMC.
- 2024.09.05. Genericity of pseudo-Anosovs and quasi-isometries. *Cornell Dynamics Seminar*.
- 2024.06.25. Sullivan's conjecture, Myrberg limit set and Hausdorff dimension. *One day workshop with Young Geometric Topologists at KAIST*.
- 2024.05.13. Continuity of escape rate of random walks on  $\text{CAT}(0)$  spaces. *Dynamics of group actions and random walks on groups*, Fields Institute.
- 2024.04.21. Confined subgroups and their growth. *Alfréd Rényi Institute*. Budapest, Hungary.
- 2024.04.08. Genericity of pseudo-Anosov mapping classes. *Young Geometric Group Theory XII*, Bristol, UK.
- 2024.04.04. Genericity of pseudo-Anosovs (and fully irreducibles). *Princeton Topology Seminar*.
- 2024.04.02. Genericity of pseudo-Anosov mapping classes. *Brandeis Topology Seminar*, Brandeis University.
- 2024.03.20. Random walks, superlinear divergence and quasi-isometry. *UBC Probability Seminar*, University of British Columbia.
- 2024.03.15. Growth of normal subgroups of a hyperbolic group. *GT GAPS Seminar*. (Online)
- 2024.02.29. Normal subgroups, confined subgroups and growth. *Korea-France Workshop on Dynamical Group Theory*, KIAS.
- 2024.01.17. Regularity of the escape rate and the asymptotic entropy of a random walk. *KAIST GT Fair 2024*, Busan, Korea.
- 2023.11.03. Genericity of contracting elements in groups. *SNU Rookies Workshop 2023*, Yangjeong, Korea.
- 2023.10.28. Generic mapping classes are pseudo-Anosov. *2023 KMS Fall Meeting*, SNU.
- 2023.10.05. Genericity of contracting elements in groups. *Tokyo-Seoul Conference in Mathematics 2023*, The University of Tokyo.
- 2023.09.11. Random walks on groups without an action on hyperbolic spaces. *Topology seminar*, KAIST.

- 2023.09.04. Genericity of contracting elements in groups. *World of GroupCraft III*. (Online)
- 2023.08.21. Genericity of contracting elements in groups. *The 4th Korea-France Conference in Mathematics, KIAS*.
- 2023.07.14. Geometry and dynamics of a finitely generated group. *Workshop on Quantum Analysis 2023, Sokcho, Korea*.
- 2023.04.20. Random walks on non-positively curved spaces: rate of escape, CLT and large deviation principle. *Conference on Probability and PDE, KAIST*.
- 2023.04.06. Counting pseudo-Anosov mapping classes. (series of 4 talks) *Virtual Seminar on Geometry and Topology, KIAS*.
- 2023.02.09. Asymmetry of a generic outer automorphism of a free group. *The 18th East Asian Conference on Geometric Topology*. (Online)
- 2022.11.08. Random subgroup is quasi-isometrically embedded. *Geometry and Topology Seminar, CUNY*.
- 2022.11.08. The log-regularity of the hitting measure on the Gromov boundary. *Group actions, Geometry and Dynamics Seminar, Yale University*.
- 2022.11.02. Asymmetry of a typical outer automorphism. *Geometry and Topology Seminar, Temple university*.
- 2022.10.27. Simple length spectrum of a hyperbolic surface. *Hyperbolic lunch, University of Toronto*.
- 2022.10.24. Mapping class group, Teichmüller space and Bers' proof of Nielsen-Thurston's classification. *Dynamics Seminar, University of Toronto*.
- 2022.05.12. Typical behavior of random mapping classes and outer automorphisms. *Fudan Topology Seminar*. (Online)
- 2022.04.11. Limit laws beyond hyperbolic spaces. *AIM Workshop "Random walks beyond hyperbolic groups", San Jose, CA United States*
- 2021.12.03. Limit laws for random walks on mapping class groups. *Conformal Dynamics and Groups Seminar, BICMR*. (Online)
- 2021.11.29. Random walks on mapping class groups favor pseudo-Anosovs. *Dynamics Seminar, University of Toronto*. (Online)
- 2021.11.18. Random walks, counting problems and genericity of loxodromics. *Geometry and Topology Seminar, Technion*. (Online)
- 2021.10.06. Random walks on Gromov hyperbolic spaces and Teichmüller spaces. *Tokyo Tech Topology Seminar*. (Online)
- 2021.07.22. Random walks on Gromov hyperbolic spaces and Teichmüller spaces. *Pacific Dynamics Seminar*. (Online)
- 2021.02.03. Simple length spectra of a generic hyperbolic surface determine its isometry class. *Virtual Seminar on Geometry and Topology*. (Online)

### Contributed Talks

- 2022.05.31. Frequently contracting geodesics and random mapping class. *PK2 Topology Workshop*.
- 2022.04.25. Limit laws and their consequences on MCG and  $\text{Out}(Fn)$ . 5-min lightning talk, *IHP Conference "Mapping Class Groups and  $\text{Out}(Fn)$ "*.
- 2022.01.18. Random mapping classes are pseudo-Anosov. *The 17th East Asian Conference on Geometric Topology*. (Online)
- 2021.01.26. Rigidity of length Identities of hyperbolic surfaces. *The 16th East Asian Conference on Geometric Topology*. (Online)
- 2020.10.30. Random walks and mapping class groups. 10-minute talk at *Geometric Group Theory in East Asia*. (Online)

## Conference organization

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- 2026.06.07. Young Geometric Group Theory (yGGT) XIV, KIAS (with Carl-Fredrik Nyberg Brodda, Yu-Chan Chang, Giovanni Italiano, Annette Karrer, Sang-hyun Kim, Sanghoon Kwak and Amelia Pompilio)
- 2025.11.26. Dynamical Group Theory VII (with KyeongRo Kim, Sang-hyun Kim, Seonhee Lim and Donggyun Seo)
- 2025.08.29. 2025 2K-GATE One day Geometric Topology Festival, KIAS (with Hyungryul Baik and Sang-hyun Kim)
- 2023.12.18–20. Hyperbolic Geometry of Numbers, KIAS (with Sang-hyun Kim and Ser-Peow Tan)
- 2023.09.27–28. The 1st Korea-Chile Workshop on Dynamical Group Theory, Incheon, Korea. (with Sang-hyun Kim)

## Teaching

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### At Cornell

- 2025S Instructor, MATH 3040 Prove It!
- 2024F Instructor, MATH 1110 Calculus I

### At KAIST

- 2022F TA, MAS102 Calculus II & MAS430 Combinatorial Topology
- 2022S TA, MAS109 Linear algebra and Applications & MAS201 DE and Applications
- 2021F TA, MAS201 DE and Applications & MAS441 Lebesgue Integration Theory
- 2021S TA, MAS109 Linear algebra and Applications & MAS331 Topology
- 2020F TA, MAS102 Calculus II & MAS441 Lebesgue Integration Theory
- 2020S Head TA, MAS201 DE and Applications
- 2019F TA, MAS201 DE and Applications & MAS441 Lebesgue Integration Theory
- 2019S TA, MAS201 DE and Applications & MAS331 Topology
- 2018F TA, MAS355 Mathematical Statistics
- 2017S, 2018S Undergraduate TA, MAS101 Calculus I

## Service & Outreach

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- Refereed for *Advances in Mathematics*, *Annales Henri Lebesgue*, *Annales de l'IHP (B)*, *Compositio Mathematica*, *Discrete & Continuous Dynamical Systems*, *Duke Mathematical Journal* (quick opinioin), *Forum of Mathematics*, *Sigma* (quick opinion), *Geometry & Topology*, *Israel Journal of Mathematics*, *Journal of the EMS*, *Journal of Topology*, *Journal of Topology and Analysis*, *Memoirs of the EMS*, *Transactions of AMS*, *MathSciNet Mathematical Reviews*.
- Organizing VISGAT.
- Organizing *Cornell Topology Seminar* for Fall 2024 and Spring 2025.
- Published a series of math cartoons on KIAS webzine  $\langle$ Horizon $\rangle$  (in Korean)